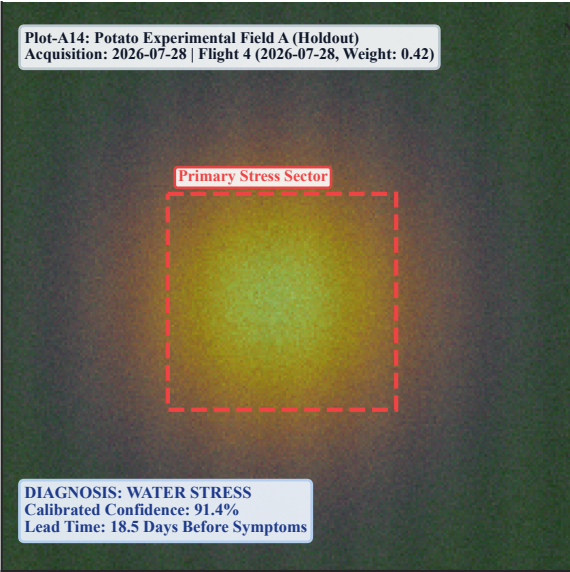
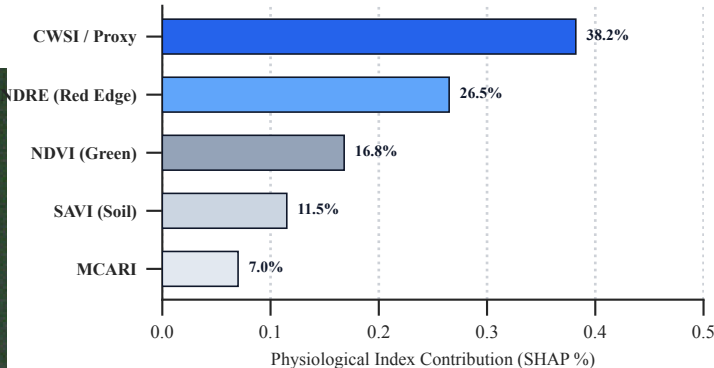


(a) Spatial Canopy Saliency (Grad-CAM Overlay)



(b) Multi-Modal Physiological Attribution



(c) Management Recommendations & Caveats

AGRONOMIC INTERVENTION PROTOCOL:

- Action: Initiate targeted deficit irrigation immediately in Plot-A14 (Canopy Sector B2). Estimated 18.5 days lead-time before irreversible chlorosis and yield loss.
- Estimated Window: 18.5 days before visible symptom onset.

MANDATORY OPERATIONAL CAVEATS:

- Thermal channel reconstructed via MS Proxy (FLIR payload offline).
- Lead-time estimate calibrated on staged ground truth (D3).
- Agronomist field scout verification recommended prior to application.